## Using TypeScript

We've already installed TypeScript back in *Adding TypeScript* by using the tsc_watch tool to run TypeScript's type checker over our code base, and then using esbuild to convert our TypeScript code into JavaScript. TypeScript's configuration is managed by a `tsconfig.json` file, which basically specifies what TypeScript allows and the kind of code that TypeScript emits. (I'll talk about more about the `tsconfig.json` file in Chapter 13, *Validating Code with Advanced TypeScript*.)

With TypeScript in place and with tsc_watch, every time we make a file change the TypeScript type checker will run, and if all its type checking passes, esbuild will convert it to JavaScript that can be sent to the browser. If the type checking does not pass, the TypeScript compiler will return error messages explaining the problem.

The important bit here is that once the code is compiled, TypeScript is done. TypeScript provides no protection at run time if the behavior of your data does not match expectations. This is usually fine as long as all the data is created by your code, but it can be a problem if your code is accepting external data that has not been type checked (for example, incoming JSON data from a server). Client-side TypeScript can guarantee that you are dealing with the data consistently in your code, but it cannot guarantee that the incoming data has the structure you expect.